

DELTA MARSH, MB, Canada

WMO: 715630

Lat: 50.183N

Lon: 98.382W

Elev: 815

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 97-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-18.8	-29.0	1.1	-22.7	-24.5	1.5	-18.7	26.0	6.0	23.0	10.2	7.0	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.7	71.2	81.5	69.0	78.7	67.4	74.5	81.8	71.8	78.6	69.4	75.9	10.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	122.0	79.5	69.3	110.8	76.5	67.0	102.3	74.0	38.6	83.0	36.1	78.8	34.0	75.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.4	19.1	-30.7	90.7	4.9	2.8	-34.2	92.7	-37.1	94.3	-39.8	95.8	-43.4	97.8		
			-30.9	78.4	4.8	3.7	-34.4	81.0	-37.2	83.1	-39.9	85.2	-43.5	87.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	4.4	8.4	20.6	39.0	51.4	61.2	67.1	64.4	55.4	41.7	26.8	10.4	(6)
	DBStd	24.16	14.07	13.90	13.06	10.02	8.23	6.63	5.24	5.98	7.36	7.81	10.77	13.10	(7)
	HDD50	6162	1415	1166	913	353	83	3	0	0	29	277	696	1227	(8)
	HDD65	10183	1880	1586	1378	781	425	150	39	85	298	723	1146	1692	(9)
	CDD50	1674	0	0	0	22	126	339	530	446	192	19	0	0	(10)
	CDD65	220	0	0	0	0	3	36	104	66	11	0	0	0	(11)
	CDH74	1978	0	0	0	9	105	323	838	559	129	15	0	0	(12)
	CDH80	439	0	0	0	0	23	80	186	129	19	2	0	0	(13)
(14) Wind	WSAvg	8.9	8.8	8.6	8.5	9.2	9.9	8.6	7.2	8.4	9.3	9.9	9.8	9.0	(14)
(15) Precipitation	PrecAvg	20.3	0.8	0.7	1.1	1.3	2.6	3.5	2.7	2.2	1.8	1.5	1.1	1.1	(15)
	PrecMax	27.6	1.9	1.8	2.6	3.1	6.0	5.7	4.7	5.1	3.9	3.3	4.6	1.9	(16)
	PrecMin	16.4	0.3	0.0	0.2	0.4	0.8	1.3	0.7	0.6	0.6	0.2	0.1	0.3	(17)
	PrecStd	2.8	0.4	0.5	0.7	0.7	1.4	1.3	1.1	1.2	0.9	0.9	1.0	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.0	55.9	75.0	83.0	87.4	88.8	88.3	82.5	74.6	57.7	40.5	(19)
		MCWB	35.6	35.9	46.1	56.7	59.4	71.8	76.3	75.1	66.8	57.6	46.9	35.8	(20)
	2%	DB	34.0	36.1	46.2	67.4	76.3	82.0	84.8	83.5	77.8	65.4	50.3	36.0	(21)
		MCWB	31.8	33.8	40.6	51.7	57.0	67.2	73.3	71.1	65.0	53.4	42.0	33.2	(22)
	5%	DB	29.7	33.1	40.9	61.6	71.0	77.3	81.5	80.0	73.3	59.2	44.5	31.3	(23)
		MCWB	28.1	31.9	37.1	49.4	56.3	64.4	70.7	69.3	63.0	50.7	39.5	29.7	(24)
	10%	DB	25.2	29.0	36.8	55.2	65.6	73.6	78.8	76.5	68.4	54.2	40.2	28.2	(25)
		MCWB	24.0	27.6	34.4	46.6	54.9	63.3	68.9	67.7	60.5	47.9	36.8	27.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	36.6	46.7	57.6	63.8	75.1	78.3	78.0	70.0	58.8	47.7	36.5	(27)
		MCDB	38.7	38.8	53.0	69.6	74.4	82.8	86.2	84.5	78.4	73.2	57.0	39.6	(28)
	2%	WB	31.9	34.0	40.5	53.3	60.3	70.3	75.3	73.7	66.7	54.8	43.0	33.4	(29)
		MCDB	33.5	35.4	45.9	65.8	69.4	77.4	82.6	81.0	74.2	62.9	48.2	35.9	(30)
	5%	WB	28.3	31.9	37.2	49.8	57.9	67.2	72.8	70.9	64.2	51.2	39.6	29.7	(31)
		MCDB	29.6	33.4	40.6	60.2	66.8	73.3	79.6	77.6	71.1	58.1	44.3	31.2	(32)
	10%	WB	24.0	27.7	34.6	47.0	55.8	64.5	70.3	68.6	61.7	48.4	37.0	27.1	(33)
		MCDB	25.1	29.1	36.7	55.3	64.2	72.2	76.7	75.0	67.5	53.4	40.2	28.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.2	18.5	19.1	18.5	17.5	19.0	19.7	20.1	17.6	15.4	15.1	(35)
		MCDBR	18.1	16.8	21.8	30.4	30.3	24.9	23.8	25.2	28.9	29.5	23.2	18.4	(36)
	5% WB	MCWBR	16.1	14.4	15.5	17.2	15.0	12.7	13.7	15.2	17.3	17.8	15.7	15.9	(37)
		MCWBR	17.9	16.3	20.5	28.1	23.2	20.3	21.9	21.5	23.9	26.7	21.6	17.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.255	0.276	0.300	0.354	0.374	0.378	0.390	0.382	0.347	0.299	0.266	0.246	(40)
	taud		2.284	2.239	2.330	2.343	2.350	2.372	2.357	2.372	2.455	2.557	2.452	2.327	(41)
	Ebn at Noon		252	274	288	282	280	279	273	270	269	268	250	240	(42)
	Edh at Noon		21	29	32	36	37	37	37	35	29	22	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		438	816	1244	1607	1758	1867	1976	1635	1180	702	385	312	(44)
	RadStd		36	55	95	110	153	131	84	119	91	91	53	33	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	-1.01	-1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page